



NAUVAL ZULFIKAR

Formal Methods in NLP · Interpretability and Efficiency of Language Models Bandung, Indonesia (relocatable to Leiden, 2026-09-01) zulfikar.nauval1998@gmail.com · +44 7300 469048 · +62 821 2567 2264 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Visa status:** UK Graduate Visa (active). Indonesian passport. Eligible for EU Researcher (Hosting Agreement) visa for European PhDs.

PROFILE

MSc Business Analytics graduate with a First Class degree from Aston University, specialising in NLP and formal methods. My dissertation, supervised by Dr. Ammar Al-Bazi, focused on blockchain-integrated LLM analysis for supply chain customer reviews, employing DeBERTa-v3 transformer fine-tuning and Ethereum smart-contract architecture. This foundation aligns with my interest in formal methods for language models, aiming to enhance interpretability and efficiency in NLP applications.

EDUCATION

Aston University, Birmingham, UK

MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**)

Ritsumeikan Asia Pacific University, Beppu, Japan

BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship
- **Distinctions:** Business Data Analysis, Consumer Behaviour, Marketing Research

RESEARCH-RELEVANT PROJECTS

LLM-Generated Adaptive Shipper Decision Rules

Solo builder, 2026

- Developed an end-to-end RAG system using gpt-4o-mini, augmented with retrieval over a curated behavioural-literature knowledge base. Focused on reproducibility and ablation studies.
- [GitHub](#)

MSc Dissertation Pipeline (DeBERTa-v3 fine-tune)

2024

- Fine-tuned DeBERTa-v3 on multi-tier supplier review data for multi-aspect classification. Evaluated emergent biases against literature baselines. Integrated into a larger decision system. Graded 1:1.

Crowd-Shipping ABM Behavioural Sweep

Solo builder, 2026

- Conducted a 450-run parameter sweep to analyse behavioural responses to reward and supply-density grids. Results anchored in literature, demonstrating significant acceptance variability.
- [GitHub](#)

VRPOD MILP Optimisation

Solo builder, 2026

- Replicated the VRPOD problem using MILP, achieving a 38.5% cost reduction on a Birmingham-inspired instance.
- [GitHub](#)

INDUSTRY EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung

Data Scientist & Project Manager · Sep 2022 – Present

- Automated ETL pipelines and developed self-service dashboards, resulting in 45% faster project delivery and 20% data-efficiency gain.

Syncwell, Birmingham, UK

Marketing Data Scientist · Jul – Sep 2024

- Implemented predictive ML and A/B testing, increasing CTR by 40% and follower growth by 22%.

PCOS Challenge, Birmingham, UK

Strategic Data Scientist Specialist · Jun – Aug 2024

- Developed a customer attribution model and automated report extraction, reducing manual workload by 80%.

Bank Muamalat Indonesia, Jakarta

Lead Data Scientist · Nov 2019 – Aug 2022

- Developed NLP models for credit portfolio analysis and risk scoring, achieving a 5.3% investment return.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of ML regressors for retail sales forecasting accuracy.

TECHNICAL SKILLS

ML/NLP: PyTorch, HuggingFace Transformers, DeBERTa-v3 fine-tuning, gpt-4o-mini API, RAG, sentence-transformers

Languages/Infrastructure: Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, FastAPI, Streamlit

Mathematical/Formal: MILP (Pyomo, scipy.optimize.milp, HiGHS), Agent-Based Modelling, Discrete-Event Simulation, Bayesian methods

Governance: Model validation, audit-grade documentation, A/B testing, AWS, Docker, Git

LANGUAGES

Indonesian: Native

English: Fluent (IELTS 7.0 overall: L7.5/R7.5/W6.5/S7.0)

Japanese: Intermediate

REFEREES

Dr. Ammar Al-Bazi

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Dr. Viktor Pekar

Lecturer, Aston Business & Social Sciences

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